

HMS 3

Hospital Management System

Comprehensive Project Report

Author: Gujjula Samara Simha Reddy

GitHub: github.com/gujjulassr/hms_3

Date: April 2026

FastAPI + Streamlit + LangGraph + PostgreSQL + MongoDB + ChromaDB

OpenAI GPT-4o-mini + Whisper ASR + TTS + RAG

Table of Contents

1. Project Overview
2. System Architecture
3. Technology Stack & Packages
4. Database Schema & Relations
5. API Endpoints (48 endpoints)
6. LangGraph Multi-Agent System
7. Services & Business Logic
8. Frontend Dashboards
9. Advanced Python Concepts
10. Key Workflows & Flowcharts
11. Testing (148 test cases)
12. External Integrations
13. Configuration & Deployment

1. Project Overview

HMS 3 is a full-stack Hospital Management System designed to handle the complete lifecycle of hospital operations including patient registration, appointment booking with slot management, doctor session management, real-time queue tracking with delay calculations, emergency handling, beneficiary management, feedback collection with RAG-powered analytics, consultation report generation, and multi-agent AI chatbot for all user roles.

The system supports 5 user roles: Patient, Doctor, Nurse, Staff, and Admin. Each role has a dedicated dashboard with role-specific features and a chat interface powered by LangGraph multi-agent system using GPT-4o-mini.

Key Features

- Appointment booking with slot management (overbooking support, lunch block)

2. System Architecture

The system follows a clean separation between backend (FastAPI REST API) and frontend (Streamlit). The backend is fully decoupled and can be consumed by any frontend (React, mobile app, etc.). All communication happens via REST API calls with JWT authentication.

Architecture Components

Component	Technology & Purpose
Frontend	Streamlit - Role-based dashboards (Patie
Backend	FastAPI - Async REST API with 48 endpoin
Database	PostgreSQL (asynpg) - 10 tables for str
Chat Store	MongoDB - Persistent chat history
Vector DB	ChromaDB - RAG feedback embeddings
AI/LLM	LangGraph + GPT-4o-mini - Multi-agent ch
Speech	OpenAI Whisper (ASR) + TTS-1 + Web Speec
Email	Gmail SMTP - Notifications + Calendar in
Storage	Google Drive - Consultation report PDFs
Auth	JWT + Google OAuth 2.0
Scheduler	Background task - Auto-complete expired

3. Technology Stack & Packages

Core Framework

Package	Why It's Needed
fastapi 0.115.0	Async web framework for REST API
uvicorn 0.30.6	ASGI server for FastAPI
streamlit 1.39.0	Web UI framework for dashboards

Database & ORM

Package	Why It's Needed
sqlalchemy 2.0.35	Async ORM for PostgreSQL
asyncpg 0.29.0	PostgreSQL async driver (non-blocking)
pymongo	MongoDB driver for chat history
chromadb	Vector database for RAG feedback

Authentication

Package	Why It's Needed
pyjwt 2.9.0	JWT token creation/validation
passlib[bcrypt] 1.7.4	Password hashing (bcrypt)
google-auth-oauthlib	Google OAuth 2.0 flow

AI / LLM / Agents

Package	Why It's Needed
langgraph 0.2.53	Graph-based multi-agent orchestration
langchain 0.3.7	Tool abstraction for LLM function callin
langchain-openai 0.2.8	OpenAI GPT-4o-mini integration
openai	Whisper ASR, TTS, embeddings, report gen

Reports & Email

Package	Why It's Needed
fpdf2	PDF generation for consultation reports
google-api-python-client	Google Drive API for report upload
smtplib (stdlib)	Gmail SMTP for email notifications

Data Science

Package	Why It's Needed
pandas >=2.0.0	Data manipulation for analytics
numpy >=1.24.0	Statistical analysis
matplotlib >=3.7.0	Chart generation

Testing

Package	Why It's Needed
pytest >=7.4.0	Unit and integration test runner
pytest-asyncio	Async test support
httpx	Async HTTP client for API testing

4. Database Schema & Relations

Entity Relationship Diagram

The database consists of 10 tables in PostgreSQL with UUID primary keys. Key relationships:

- users (1) --> (1) patients/doctors (role-based extension)
- doctors (1) --> (N) sessions (one per day)
- sessions (1) --> (N) appointments
- patients (1) --> (N) appointments
- patients (1) --> (N) beneficiaries
- patients (1) --> (N) ratings
- appointments (1) --> (1) consultation_reports
- users (1) --> (N) audit_logs

Table: users

Central user account for all roles

Column	Type	Description
id	UUID PK	Primary key
email	String UNIQUE	Login email, indexed
password_hash	String	Bcrypt hash
full_name	String	Display name
phone	String	Contact number
role	String	patient/doctor/nurse/staff/admin
is_active	Boolean	Account active status
created_at	DateTime	Registration timestamp

Table: patients

Patient-specific profile data

Column	Type	Description
id	UUID PK	Primary key
user_id	UUID FK->users	Links to user account
uhid	String UNIQUE	Universal Health ID (HMS-YYYY-NNNNN)
gender	String	Male/Female/Other
blood_group	String	A+/A-/B+/B-/O+/O-/AB+/AB-
date_of_birth	Date	Patient DOB
address	String	Address
emergency_contact_name	String	Emergency contact person
emergency_contact_phone	String	Emergency phone
risk_score	Integer	0=good, +10 cancel, +20 no-show

Table: doctors

Doctor-specific profile

Column	Type	Description
id	UUID PK	Primary key
user_id	UUID FK->users	Links to user account
specialization	String	General Medicine, Cardiology, etc.
max_patients_per_day	Integer	Default 30

Table: sessions

Doctor daily sessions (one per day)

Column	Type	Description
id	UUID PK	Primary key
doctor_id	UUID FK->doctors	Which doctor
session_date	Date	Session date (indexed)
start_time	Time	e.g. 09:00
end_time	Time	e.g. 17:00
lunch_start	Time	Default 13:00 (auto-blocked)
lunch_end	Time	Default 14:00 (1hr lunch)
slot_duration_minutes	Integer	Default 15
max_per_slot	Integer	Default 2 (overbooking)
total_slots	Integer	Auto-calculated excl. lunch
status	String	scheduled/active/completed/cancelled
delay_minutes	Integer	Dynamic delay tracking
overtime_minutes	Integer	Extension beyond end_time

Table: appointments

Patient appointments with full lifecycle

Column	Type	Description
id	UUID PK	Primary key
session_id	UUID FK->sessions	Which session
patient_id	UUID FK->patients	Which patient
booked_by	UUID FK->users	Who booked
slot_number	Integer	0=emergency, 1+=normal
slot_position	Integer	Position within slot
slot_time	Time	Scheduled time
status	String	booked/checked_in/in_progress/completed/

priority	String	NORMAL/HIGH/CRITICAL
is_emergency	Boolean	Emergency flag
checked_in_at	DateTime	Patient arrival time
called_at	DateTime	Doctor called time
started_at	DateTime	Consultation start
completed_at	DateTime	Consultation end
notes	String	Visit reason / doctor notes

Table: beneficiaries

Patient family members

Column	Type	Description
id	UUID PK	Primary key
patient_id	UUID FK->patients	Parent patient
name	String	Family member name
relationship	String	Spouse/Child/Parent/Sibling
phone	String	Contact
gender	String	Gender
blood_group	String	Blood group
date_of_birth	Date	DOB

Table: ratings

Patient feedback for doctors

Column	Type	Description
id	UUID PK	Primary key
patient_id	UUID FK->patients	Who rated
doctor_id	UUID FK->doctors	Who was rated
rating	Integer	1-5 stars
feedback	Text	Free text feedback
created_at	DateTime	Submission time

Table: consultation_reports

LLM-generated reports

Column	Type	Description
id	UUID PK	Primary key
appointment_id	UUID FK->appointments	Which appointment
doctor_id	UUID FK->doctors	Which doctor
patient_id	UUID FK->patients	Which patient
content	Text	LLM-generated report content
doctor_notes	Text	Doctor's input notes
drive_link	String	Google Drive shareable URL
pdf_path	String	Local PDF file path
created_at	DateTime	Generation time

Table: audit_logs

Full action audit trail

Column	Type	Description
id	UUID PK	Primary key
user_id	UUID FK->users	Who performed action
action	String	BOOK/CANCEL/CHECKIN/CALL/COMPLETE/EMERGE
target_type	String	appointment/patient/session/beneficiary
target_id	UUID	Target record ID
details	JSON	Context: UHID, doctor, times, etc.
created_at	DateTime	Action timestamp

5. API Endpoints (48 Total)

Authentication (/api/auth) - 6 endpoints

Method	Path	Description
POST	/register	Register patient/doctor/staff with UHID
POST	/login	Email/password login, returns JWT
POST	/google-form-register	Google Form auto-register + book
POST	/change-password/{email}	Change password
GET	/google/login	Google OAuth redirect URL
GET	/google/callback	OAuth callback, auto-register

Patient (/api) - 14 endpoints

Method	Path	Description
GET	/my-profile	Patient full profile
PUT	/my-profile	Update profile details
GET	/my-appointments	Appointments (self + beneficiaries)
GET	/my-reports	Consultation reports with Drive links
GET	/doctors	List/filter doctors by specialization
GET	/available-slots	Available slots for doctor+date
POST	/book-appointment	Two-step booking with confirmation
POST	/cancel-my-appointment	Cancel (risk +10)
POST	/reschedule-appointment	Reschedule (no penalty)
GET	/my-beneficiaries	List beneficiaries
POST	/my-beneficiaries	Add beneficiary (auto-registers)
PUT	/my-beneficiaries/{id}	Update beneficiary
DELETE	/my-beneficiaries/{id}	Delete beneficiary
GET	/appointments	Query with filters

Doctor (/api/doctor) - 12 endpoints

Method	Path	Description
GET	/my-sessions	Upcoming sessions
POST	/create-session	Create session (1 per day)
POST	/activate-session	Activate (today only)
POST	/complete-session	Complete + no-show handling
POST	/extend-session	Add overtime minutes
POST	/cancel-session	Cancel session

GET	/queue	Real-time queue (4 categories)
POST	/checkin-patient	Check in (booked->checked_in)
POST	/call-patient	Call (checked_in->in_progress)
POST	/complete-appointment	Complete + report generation
POST	/cancel-appointment	Cancel appointment
POST	/emergency-book	Emergency (bypasses slots)

Chat & Voice (/api/chat) - 6 endpoints

Method	Path	Description
POST	/message	Text chat via LangGraph agent
POST	/voice	Audio->Whisper->LLM->TTS->Audio
POST	/speak	Text to speech (OpenAI TTS)
POST	/transcribe	Audio to text (Whisper)
GET	/history	Chat history from MongoDB
DELETE	/history	Clear chat history

Admin (/api/admin) - 11 endpoints

Method	Path	Description
GET	/stats	System overview metrics
GET	/users	All users with full details
PUT	/users/{id}	Edit any user (admin only)
POST	/toggle-user/{id}	Activate/deactivate (admin)
GET	/doctors	Doctors with session/appt stats
GET	/patients	Patients with risk scores
GET	/sessions	All sessions with filters
POST	/sessions/{id}/activate	Admin activate session
POST	/sessions/{id}/cancel	Admin cancel session
POST	/sessions/{id}/complete	Admin complete session
GET	/audit-logs	Full audit trail

Analytics (/api/analytics) - 5 endpoints

Method	Path	Description
GET	/feedback-rag	RAG semantic feedback search
POST	/feedback-sync	Sync ratings to ChromaDB
GET	/report	Full analytics dashboard
GET	/busiest-doctors	Doctor rankings
GET	/peak-hours	Peak hour analysis

6. LangGraph Multi-Agent System

The chat system uses LangGraph to create role-specific AI agents. Each agent has access to different tools based on the user's role. The supervisor graph routes messages to the correct agent based on the JWT token's role claim. All agents use GPT-4o-mini with function calling.

Agent Architecture

User Message -> Supervisor Graph -> Role Router

Patient -> Patient Agent (15 tools)

Doctor -> Doctor Agent (21 tools)

Staff/Admin -> Staff Agent (35 tools)

Each agent: LLM decides tool -> Tool executes (DB query) -> LLM formats response -> Save to MongoDB -> Return to user

Agent	Tools	Capabilities
Patient Agent	15	Book/cancel/reschedule appointments, sea
Doctor Agent	21	Queue management, sessions, emergency bo
Staff/Admin Agent	35	Everything: registration, booking, queue

Key Agent Rules

- Always use tools for real-time data (never reuse old chat context)

7. Services & Business Logic

Notification Service

Gmail SMTP emails with calendar invites (.ics) for booking, cancellation, reschedule, check-in, no-show, feedback request, session cancelled

Report Generator

LLM generates consultation report -> FPDF creates PDF -> Google Drive upload -> Email with PDF attachment + Drive link

RAG Feedback

ChromaDB vector store + OpenAI text-embedding-3-small. Semantic search on patient feedback. Generates insightful summaries with themes and patterns

Speech Service

OpenAI Whisper for ASR (speech-to-text), TTS-1 for text-to-speech (6 voice options: alloy, echo, fable, onyx, nova, shimmer)

Chat Store

MongoDB persistence for chat history. Save/retrieve/clear per user email

Scheduler

Background async task (120s interval). Auto-completes expired sessions: booked->no_show (+20 risk), checked_in->cancelled (no penalty)

Slot Utils

Generates time slots from start to end, skipping lunch break (13:00-14:00). Counts slots, checks lunch time

Audit Logger

Structured audit trail with actor, action, target, details JSON. Used across all endpoints

8. Frontend Dashboards (Streamlit)

Patient Dashboard (6 tabs)

- My Details - View/edit profile, emergency contact, beneficiaries

Doctor Dashboard (4 tabs)

- Queue - Real-time queue (emergency/in-progress/waiting/booked), check-in/call/complete, emergency booking

Admin Dashboard (6 tabs)

- Overview - System stats (users, patients, doctors, appointments, no-shows)

9. Advanced Python Concepts

Object-Oriented Programming

Abstract Base Classes (ABC), inheritance (Person->Patient/Doctor), encapsulation (private attrs + @property), polymorphism (overridden methods), magic methods (`__repr__`, `__eq__`, `__hash__`, `__lt__`, `__len__`, `__iter__`)

Data Structures & Algorithms

Binary search for $O(\log n)$ slot lookup (ScheduleManager), priority queue sorting (emergency first), Timsort with composite lambda keys, thread-safe per-slot locking

Functional Programming

List comprehensions (available_slots, active_appointments), generators (slot_generator, waiting_generator), lambda functions (sorting keys), map/filter (get_slot_times, get_emergency_patients)

Decorators

@log_action (logs function calls), @timer (execution time), @require_role (access control), @retry (auto-retry with backoff). All support async/sync

Threading & Concurrency

Per-slot thread locks (prevent double-booking), ThreadPoolExecutor (concurrent booking), run_async_in_thread bridge. Production uses async/await (asyncpg)

Regex Validation

Compiled patterns for email, phone, UHID (HMS-YYYY-NNNNN), 24h time, date (YYYY-MM-DD), name, password strength, blood group. Text extraction utilities

Async/Await

Full async stack: asyncpg (DB), FastAPI (endpoints), LangGraph (agents), background scheduler (asyncio.sleep loop)

10. Key Workflows

Appointment Booking

Patient selects doctor+date -> View available slots -> Click slot -> Confirm -> Book (duplicate check) -> Email + calendar invite -> Audit log

Session Lifecycle

Create (scheduled) -> Activate (today only) -> Patients: Book->Check-in->Call->Complete (delay updated) -> Session Complete: booked->no_show(+20), checked_in->cancelled(0)

Rescheduling

Patient requests reschedule -> Check new slot availability -> If full: suggest next available -> Old appointment: status=rescheduled (no risk penalty) -> New appointment: status=booked -> Email notification

Emergency

Doctor/Staff enters UHID -> Create emergency appointment (slot=0, priority=CRITICAL) -> Auto checked-in -> Shows above normal queue -> Doctor calls -> Complete

Consultation Report

Doctor completes appointment -> LLM generates report (diagnosis, prescription, follow-up) -> PDF created -> Google Drive upload -> Email to patient with Drive link + PDF

Google Form Pipeline

Patient fills Google Form -> Apps Script triggers -> POST /api/auth/google-form-register -> Validate -> Register (if new) -> Book appointment -> Email confirmation

RAG Feedback

Patient submits rating -> Auto-synced to ChromaDB -> Admin asks question -> Semantic search -> Retrieve relevant feedback -> LLM generates summary with themes and patterns

Risk Score

Cancel: +10, No-show: +20, Reschedule: 0, Check-in not seen: 0 (hospital fault). Displayed in admin dashboard (color-coded: green/orange/red)

11. Testing (148 Test Cases)

End-to-End Tests (81 - bash)

Category	Tests	Coverage
Authentication	10	Login, register, duplicate, wrong passwo
Patient Profile	5	Get/update profile, verify changes
Doctor Sessions	7	Create, duplicate, past date, activate,
Appointment Booking	8	Book, duplicate, overbooking, full slot,
Queue Flow	7	Check-in, call, complete, double check-i
Cancel	3	Cancel booked, already cancelled, patien
Reschedule	4	Free slot, full slot block, no risk pena
Emergency	4	Book, queue, duplicate, non-existent
Beneficiaries	5	Add, get, auto-register, update, delete
Session Completion	4	Complete, no-show, risk score, cancelled
Admin	12	Stats, users, filter, doctors, patients,
Doctors & Slots	3	List, filter, slots
Chat History	4	Send, history, clear, empty
Edge Cases	5	Health, root, unauthorized, invalid toke

Unit Tests (67 - pytest)

Module	Tests	Coverage
OOP Classes	39	Person hierarchy, magic methods, polymor
Validators	15	Email, phone, UHID, time, date, name, pa
Slot Utils	13	Slot generation, lunch block, overtime c

12. External Integrations

Integration	Purpose
OpenAI GPT-4o-mini	LLM for chat agents and report generatio
OpenAI Whisper	Speech-to-text (ASR) for voice chat
OpenAI TTS-1	Text-to-speech with 6 voice options
OpenAI Embeddings	text-embedding-3-small for RAG vectors
Google OAuth 2.0	Patient auto-registration via Google log
Google Drive API	Upload consultation report PDFs, shareab
Google Forms	External registration + auto-booking pip
Gmail SMTP	Email notifications with calendar invite
ChromaDB	Persistent vector database for RAG feedb
MongoDB	Chat history persistence across sessions
Web Speech API	Browser-native continuous speech recogni

13. Configuration & Deployment

Environment Variables (.env)

Variable	Purpose
DATABASE_URL	PostgreSQL connection string
JWT_SECRET	JWT signing secret
JWT_ALGORITHM	HS256
JWT_EXPIRY_MINUTES	Token expiry (480 = 8 hours)
OPENAI_API_KEY	OpenAI API key for all AI features
LLM_MODEL	gpt-4o-mini
SMTP_HOST/PORT/USER/PASSWORD	Gmail SMTP for emails
GOOGLE_CLIENT_ID/SECRET	Google OAuth credentials
GOOGLE_REDIRECT_URI	OAuth callback URL
GOOGLE_DRIVE_REFRESH_TOKEN	Drive API access
MONGO_URL/DB	MongoDB connection for chat
PUBLIC_REGISTER_KEY	API key for Google Form endpoint

How to Run

- 1. Clone: `git clone github.com/gujjulasr/hms_3.git`